Analysing Sectoral Employment Dynamics in the U.S. Using Autoregressive Models for Matrix-Valued Time Series

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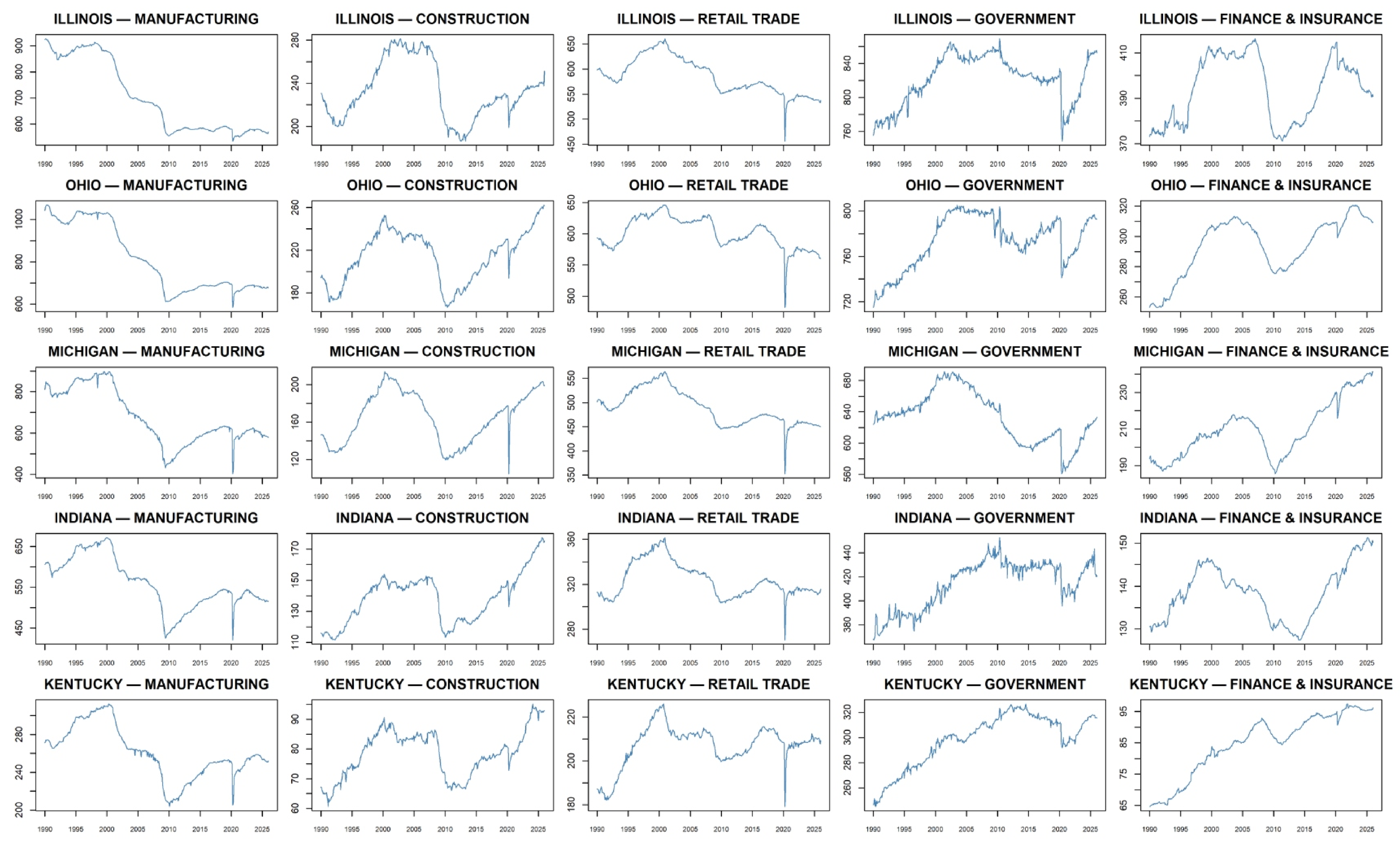
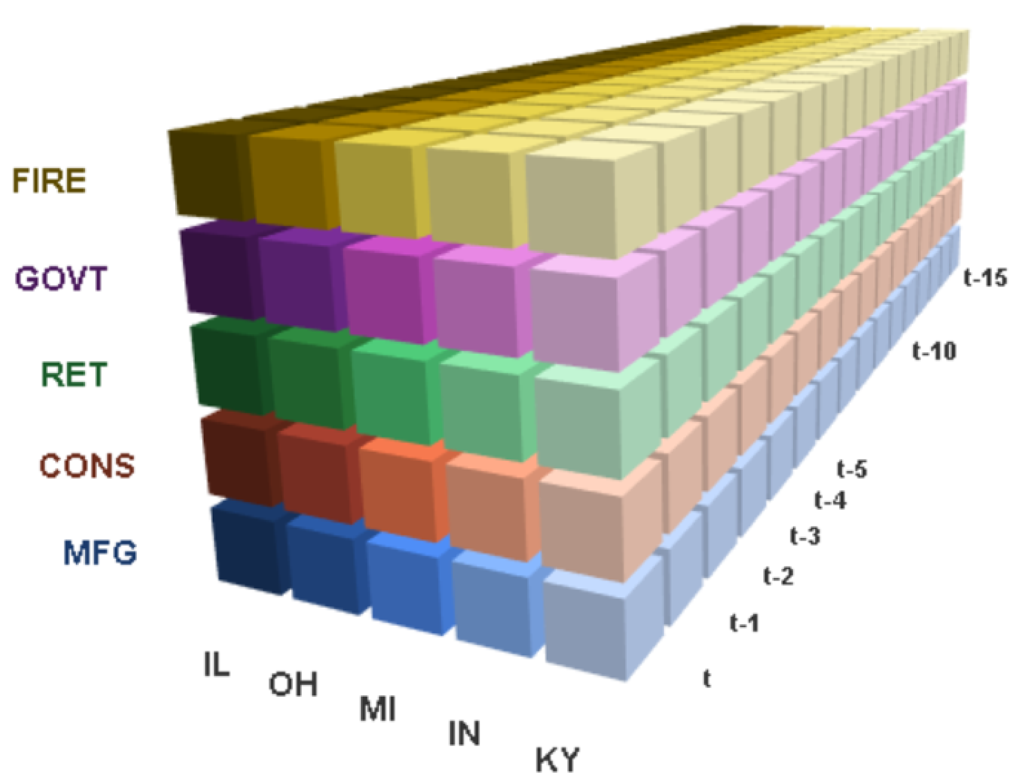
Motivation & Research Question

Employment shocks rarely stay in one place. A slowdown in **manufacturing** can spill into **construction** and **retail**, while a downturn in one state spreads to its neighbours through supply chains, trade, and migration. Regional employment is therefore *two-dimensional* — it has a **sector** dimension and a **state** dimension — observed **over time**. Each month is naturally a **matrix**, not a flat list of numbers.



Raw sectoral employment levels: 5 states $\times$ 5 sectors, 1990–2026.

Stacking these 5×5 matrices month after month yields a **three-dimensional tensor** — state $\times$ sector $\times$ time:

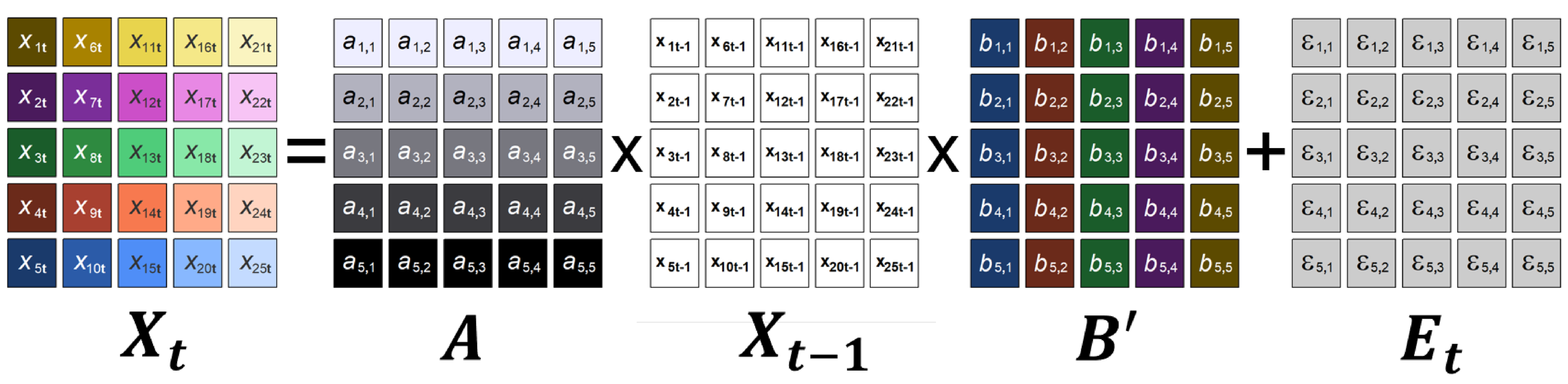


Research question

Can a **Matrix Autoregression (MAR)** capture how employment dynamics propagate *jointly* across states and sectors — **parsimoniously**, **interpretably**, and forecasting at least as well as a standard VAR?

The MAR Model

$$\boldsymbol{X}_t = \boldsymbol{A}\,\boldsymbol{X}_{t-1}\,\boldsymbol{B}' + \boldsymbol{E}_t \iff \operatorname{vec}(\boldsymbol{X}_t) = (\boldsymbol{B} \otimes \boldsymbol{A})\operatorname{vec}(\boldsymbol{X}_{t-1}) + \operatorname{vec}(\boldsymbol{E}_t)$$



The MAR model separates temporal dependence into a row effect and a column effect, so interactions in each dimension can be modeled parsimoniously.

- A **(left)** — captures dependence across *states* (columns): how shocks in one state affect other states over time.
- B **(right)** — captures dependence across *sectors* (rows): how shocks in one sector affect other sectors over time.

Why Not Just Use a VAR?

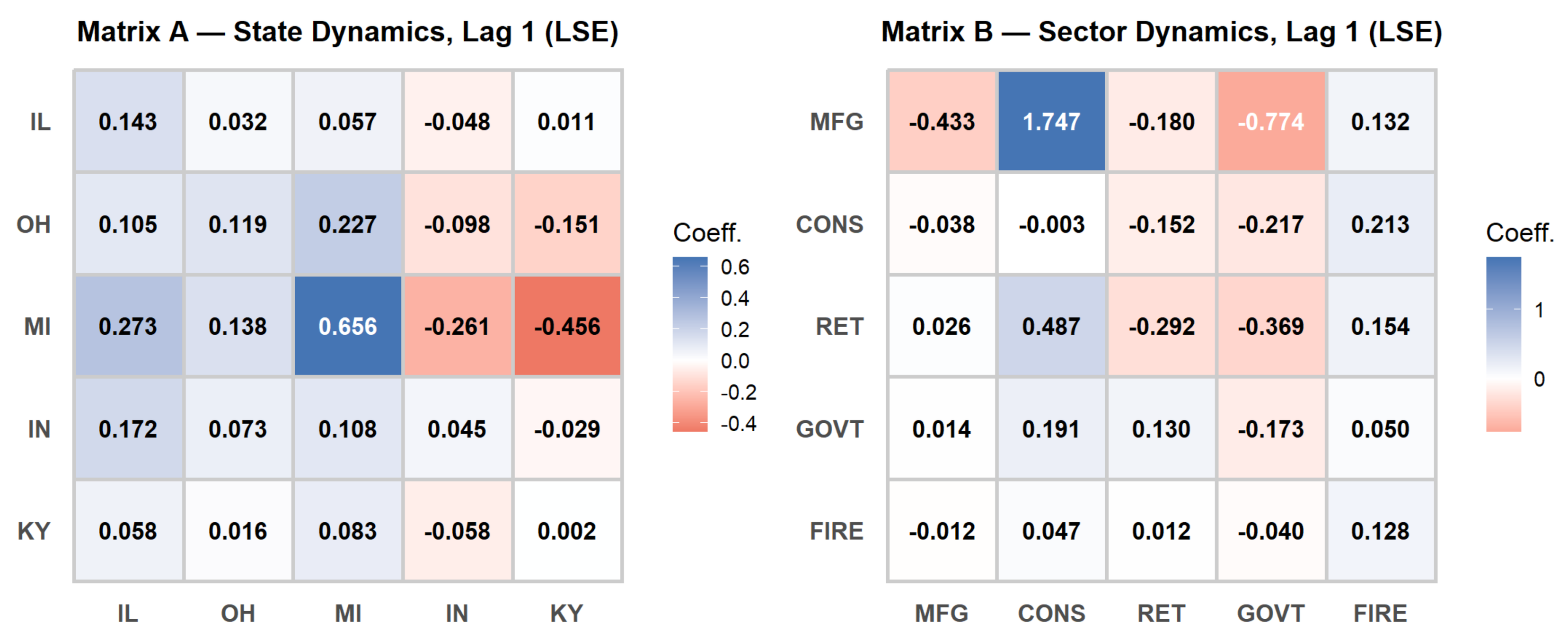
A standard VAR must **vectorise** every matrix — stacking all its entries into one long column — which **throws away the row/column structure** that makes the data interpretable. It also explodes in size: VAR(1) needs $m^2n^2 = \mathbf{625}$ coefficients, while MAR(1) needs only $m^2+n^2 = \mathbf{50}$ — about **8%** as many.

Data Preparation

- Raw employment is **non-stationary** (ADF $p \in [0.55, 0.96]$).
- Thus we apply **first-differencing** $\Delta y_t = y_t - y_{t-1}$, making every series stationary ($p < 0.01$).
- The core mechanism is still **autoregression**, so we select a lag order.
- **BIC** tells us to use **MAR(2)** (best in-sample fit).

P	Parameters	BIC
1	50	1.5218
2	**100**	**1.5131**
3	150	1.5136

Results (LSE, Lag 1)



- **Michigan persistence:** $A_{\text{MI,MI}} = 0.656$ — its concentrated auto/industry base keeps shocks alive over several months, and spills to OH, IN, IL.
- **Construction leads:** $B_{\text{MFG,CONS}} = 1.747$ and $B_{\text{RET,CONS}} = 0.487$ — construction predicts next-period manufacturing and retail.

Does It Forecast? MAR vs. VAR

Backtest strategy:

- Train on a **fixed window** of 302 months.
- Make a **one-step-ahead** forecast, then roll forward by one month.
- Repeat $\Rightarrow$ **129 out-of-sample windows**.
- Score accuracy by **MSFE** (mean squared forecast error).

In-sample fit

Model	Parameters	RSS
MAR(1)	49	220,318
MAR(2)	98	210,533
VAR(1)	625	196,922

Out-of-sample MSFE

Model	MSFE	Ratio
MAR(1)	**1,703**	**0.936**
MAR(2)	2,033	1.117
VAR(1)	1,820	1.000

Key result

MAR(1) beats VAR(1) out-of-sample ($\approx 6.4\%$ smaller errors) using only **50 of VAR's 625 parameters**. Parsimony wins where it counts — forecasting.

Conclusion

- MAR keeps the state $\times$ sector structure a VAR discards, with a **fraction of the parameters**.
- Coefficients are **economically interpretable** (Michigan persistence, construction leading).
- The simpler **MAR(1) forecasts best**, beating VAR(1).

Reference: Chen, R., Xiao, H., & Yang, D. (2021). Autoregressive models for matrix-valued time series. *Journal of Econometrics*, 222(2), 539–560.

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